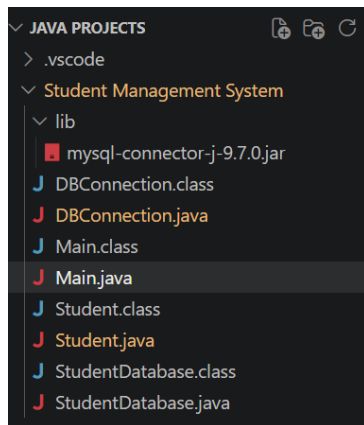




// DBConnection.java

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {

    private static final String URL = "jdbc:mysql://localhost:3306/studentdb";
    private static final String USER = "root";
    private static final String PASSWORD = "pranali";

    public static Connection getConnection() {
        Connection con = null;
        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            con = DriverManager.getConnection(URL, USER, PASSWORD);

        } catch (ClassNotFoundException e) {
            System.out.println("MySQL Driver NOT FOUND!");
            e.printStackTrace();
        } catch (SQLException e) {
            System.out.println("Connection Failed!");
            e.printStackTrace();
        }

        return con;
    }
}
```

//Main.java

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        StudentDatabase dao = new StudentDatabase();

        while (true) {

            System.out.println("\n===== STUDENT MANAGEMENT SYSTEM =====");
            System.out.println("1. Add Student");
            System.out.println("2. View Students");
            System.out.println("3. Search Student");
            System.out.println("4. Update Student");
            System.out.println("5. Delete Student");
            System.out.println("6. Exit");
            System.out.print("Enter your choice: ");

            int choice = sc.nextInt();

            switch (choice) {

                case 1:
                    System.out.print("Enter ID: ");
                    int id = sc.nextInt();
                    sc.nextLine();

                    System.out.print("Enter Name: ");
                    String name = sc.nextLine();

                    System.out.print("Enter Age: ");
                    int age = sc.nextInt();
                    sc.nextLine();

                    System.out.print("Enter Course: ");
                    String course = sc.nextLine();

                    Student student = new Student(id, name, age, course);
                    dao.addStudent(student);
                    break;

                case 2:
                    dao.viewStudents();
                    break;

                case 3:
                    System.out.print("Enter Student ID: ");
                    id = sc.nextInt();
                    dao.searchStudent(id);
                    break;

            }

        }

    }

}
```

```

case 4:
    System.out.print("Enter Student ID: ");
    id = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter New Name: ");
    name = sc.nextLine();

    System.out.print("Enter New Age: ");
    age = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter New Course: ");
    course = sc.nextLine();

    Student updateStudent = new Student(id, name, age, course);
    dao.updateStudent(updateStudent);
    break;

case 5:
    System.out.print("Enter Student ID: ");
    id = sc.nextInt();
    dao.deleteStudent(id);
    break;

case 6:
    System.out.println("Thank You!");
    sc.close();
    System.exit(0);

default:
    System.out.println("Invalid Choice!");
}
}
}
}
}

```

[// StudentDatabase.java](#)

```
import java.sql.*;
```

```
public class StudentDatabase {
```

```
    // Add Student
```

```
    public void addStudent(Student student) {
```

```
        String sql = "INSERT INTO students(id,name,age,course) VALUES(?,?,?,?)";
```

```
        try (Connection con = DBConnection.getConnection();
            PreparedStatement ps = con.prepareStatement(sql)) {
```

```
            ps.setInt(1, student.getId());
            ps.setString(2, student.getName());
            ps.setInt(3, student.getAge());
            ps.setString(4, student.getCourse());
```

```

        int rows = ps.executeUpdate();

        if (rows > 0) {
            System.out.println("Student Added Successfully!");
        }

    } catch (SQLException e) {
        System.out.println(e.getMessage());
    }
}

// View Students
public void viewStudents() {

    String sql = "SELECT * FROM students";

    try (Connection con = DBConnection.getConnection();
        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery(sql)) {

        while (rs.next()) {

            System.out.println("-----");
            System.out.println("ID : " + rs.getInt("id"));
            System.out.println("Name : " + rs.getString("name"));
            System.out.println("Age : " + rs.getInt("age"));
            System.out.println("Course : " + rs.getString("course"));
        }

    } catch (SQLException e) {
        System.out.println(e.getMessage());
    }
}

// Search Student
public void searchStudent(int id) {

    String sql = "SELECT * FROM students WHERE id=?";

    try (Connection con = DBConnection.getConnection();
        PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setInt(1, id);

        ResultSet rs = ps.executeQuery();

        if (rs.next()) {

            System.out.println("ID : " + rs.getInt("id"));
            System.out.println("Name : " + rs.getString("name"));
            System.out.println("Age : " + rs.getInt("age"));

```

```

        System.out.println("Course : " + rs.getString("course"));

    } else {
        System.out.println("Student Not Found!");
    }

} catch (SQLException e) {
    System.out.println(e.getMessage());
}
}

// Update Student
public void updateStudent(Student student) {

    String sql = "UPDATE students SET name=?, age=?, course=? WHERE id=?";

    try (Connection con = DBConnection.getConnection();
        PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setString(1, student.getName());
        ps.setInt(2, student.getAge());
        ps.setString(3, student.getCourse());
        ps.setInt(4, student.getId());

        int rows = ps.executeUpdate();

        if (rows > 0) {
            System.out.println("Student Updated Successfully!");
        } else {
            System.out.println("Student Not Found!");
        }

    } catch (SQLException e) {
        System.out.println(e.getMessage());
    }
}

// Delete Student
public void deleteStudent(int id) {

    String sql = "DELETE FROM students WHERE id=?";

    try (Connection con = DBConnection.getConnection();
        PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setInt(1, id);

        int rows = ps.executeUpdate();

        if (rows > 0) {
            System.out.println("Student Deleted Successfully!");
        }
    }
}

```

```
        } else {  
            System.out.println("Student Not Found!");  
        }  
  
    } catch (SQLException e) {  
        System.out.println(e.getMessage());  
    }  
}  
}
```

//Student.java

```
public class Student {  
  
    private int id;  
    private String name;  
    private int age;  
    private String course;  
  
    // Constructor  
    public Student(int id, String name, int age, String course)  
    {  
        this.id = id;  
        this.name = name;  
        this.age = age;  
        this.course = course;  
    }  
  
    // Getters  
    public int getId() {  
        return id;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public int getAge() {  
        return age;  
    }  
  
    public String getCourse() {  
        return course;  
    }  
  
    // Setters  
    public void setName(String name) {  
        this.name = name;  
    }  
  
    public void setAge(int age) {  
        this.age = age;  
    }  
  
    public void setCourse(String course) {
```

```

        this.course = course;
    }

    @Override
    public String toString() {
        return "ID : " + id +
            "\nName : " + name + "\nAge : " + age + "\nCourse : " + course;
    }
}

```

OUTPUT:- //1. Add Student

```

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 1
Enter ID: 5
Enter Name: Sarika
Enter Age: 25
Enter Course: BSC
Database Connected Successfully!
Student Added Successfully!

```

//2. View Students

```

1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 2
Database Connected Successfully!
-----
ID : 1
Name : Pranali
Age : 22
Course : MCA
-----
ID : 2
Name : Kajal
Age : 23
Course : MSC
-----
ID : 3
Name : Ganesh
Age : 23
Course : MCA
-----
ID : 4
Name : Ketaki
Age : 21
Course : Arch
-----
ID : 5
Name : Sarika
Age : 25

```

// 3.Search Student

```

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 3
Enter Student ID: 3
Database Connected Successfully!
ID : 3
Name : Ganesh
Age : 23
Course : MCA

```

//4. Update Student

```
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 4
Enter Student ID: 6
Enter New Name: Mahek
Enter New Age: 22
Enter New Course: MBA
Database Connected Successfully!
Student Updated Successfully!
```

// 5.Delete Student

```
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 5
Enter Student ID: 6
Database Connected Successfully!
Student Deleted Successfully!
```

//SQL

```
CREATE DATABASE studentdb;
USE studentdb;
```

```
CREATE TABLE students (
    id INT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    age INT,
    course VARCHAR(50)
);
```

```
SELECT * FROM students;
```

// Saved Database of Student

	id	name	age	course
▶	1	Pranali	22	MCA
	2	kajal	23	MSC
	3	Ganesh	23	MCA
	4	Ketaki	21	Arch
	5	Sarika	25	BSC
•	NULL	NULL	NULL	NULL

